AUTOMATING DATA EXPORT AND FORMATTING IN SQL SERVER MANAGEMENT STUDIO

SSMS

AGENDA

- ✓ Problem statement
- ✓ Solution overview
- ✓ Code Explanation
- ✓ Detailed Code Walkthrough
- ✓ Benefits
- ✓ Conclusion

MAIN CONCERNS OF

PROBLEM STATEMENT

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- **Challenge:**

  Manually exporting and formatting data from SQL Server can be time-consuming and prone to errors.

- **Impact:**

  Inefficiencies in data handling can lead to delays and inaccuracies in reporting.

SOLUTION OVERVIEW

WHAT CAN WE DO?

SOLUTION OVERVIEW

- ✓ **Objective:**

  Automate the process of exporting data from SQL Server to Excel and apply formatting to improve readability.

- ✓ **Approach:**

  Use Python with libraries to streamline the workflow.

WHAT DID WE DO!

CODE EXPLANATION

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1. Establish a connection to the SQL Server using SQLAlchemy.
2. Execute a SQL query to fetch the required data.
3. Export the data to an Excel file using pandas.
4. Load the Excel file and apply formatting using openpyxl.

DETAILED CODE

WALKTHROUGH

CODE INSIGHTS

Connection String: Define the connection string for SQL Server.

SQL Query: Write a query to select and format the data.

Data Export: Use pandas to export the query results to an Excel file.

Formatting: Apply borders and adjust column widths using openpyxl.

EXPORTED DATA

The exported content is an Excel file named UserQueryResults.xlsx. It contains user access information from a SQL Server database. The data includes the following model.

PIN	SECONDARY ID	ACCESS GROUP	ACCOUNT STATUS
User001	User001	Web Access	**ACTIVE**
User002	John	Admin Access	**ACTIVE**
User003	Martin	Auditor Access	**ACTIVE**
User004	User004	Web Access	**ACTIVE**
User005	Laura	Web Access	**ACTIVE**

BENEFITS OF THIS AUTOMATION

- ✓ **Efficiency:** Automates repetitive tasks, saving time.
- ✓ **Accuracy:** Reduces the risk of human error in data handling.
- ✓ **Consistency:** Ensures data is consistently formatted every time.
- ✓ **Readability:** Improves the presentation of data in Excel.
- ✓ **Scalability:** Easily handles large datasets and can grow with your needs.

THANK YOU

Automation is not about replacing people - it’s about empowering them to do more meaningful work.

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